CONCEPT 38.3

People modify crops by breeding and genetic engineering *(pp. 836–840)*

- Hybridization of different varieties and even species of plants is common in nature and has been used by breeders, ancient and modern, to introduce new genes into crops. After two plants are successfully hybridized, plant breeders select those progeny that have the desired traits.
- In genetic engineering, genes from unrelated organisms are incorporated into plants. Genetically modified (GM) plants can increase the quality and quantity of food worldwide and may also become increasingly important as biofuels.
- There are concerns about the unknown risks of releasing GM organisms into the environment, but the potential benefits of crops that express **transgenes** will need to be considered.

? *Give two examples of how genetic engineering has improved or might potentially improve food quality.*

TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. A fruit is
(A) a mature ovary.
(B) a mature ovule.
(C) a seed plus its integuments.
(D) an enlarged embryo sac.

2. Double fertilization means that
(A) flowers must be pollinated twice to yield fruits and seeds.
(B) every egg must receive two sperm to produce an embryo.
(C) one sperm is needed to fertilize the egg, and a second sperm is needed to fertilize the polar nuclei.
(D) every sperm has two nuclei.

3. *Bt* maize
(A) is resistant to various herbicides, making it practical to weed maize fields with those herbicides.
(B) contains transgenes that increase vitamin A content.
(C) includes bacterial genes that produce a toxin that reduces damage from insect pests.
(D) is a "boron (B)-tolerant" transgenic variety of maize.

4. Which statement concerning grafting is correct?
(A) Stocks and scions refer to twigs of different species.
(B) Stocks and scions must come from unrelated species.
(C) Stocks provide root systems for grafting.
(D) Grafting creates new species.

Levels 3-4: Applying/Analyzing

5. Some plant species produce male and female individuals in which the males have an XY genotype and the females have an XX genotype. After double fertilization, what would be the genotypes of the embryos and endosperm nuclei?
(A) embryo XY/endosperm XXX or embryo XX/endosperm XXY
(B) embryo XX/endosperm XX or embryo XY/endosperm XY
(C) embryo XX/endosperm XXX or embryo XY/endosperm XYY
(D) embryo XX/endosperm XXX or embryo XY/endosperm XXY

6. In a species showing sporophytic incompatibility, which type(s) of pollen could successfully fertilize an S_2S_3 flower?
(A) S_1 pollen from an S_1S_3 flower
(B) S_2 or S_3 pollen from an S_2S_3 flower
(C) S_3 pollen from an S_1S_1 flower
(D) S_1 pollen from an S_1S_1 flower

7. The black dots that cover strawberries are actually fruits formed from the separate carpels of a single flower. The fleshy and tasty portion of a strawberry derives from the receptacle of a flower with many separate carpels. Therefore, a strawberry is
(A) a simple fruit with many seeds.
(B) both a multiple fruit and an accessory fruit.
(C) both a simple fruit and an aggregate fruit.
(D) both an aggregate fruit and an accessory fruit.

8. **DRAW IT** Draw and label the parts of a flower.

Levels 5-6: Evaluating/Creating

9. **EVOLUTION CONNECTION** With respect to sexual reproduction, some plant species are fully self-fertile, others are fully self-incompatible, and some exhibit a "mixed strategy" with partial self-incompatibility. These reproductive strategies differ in their implications for evolutionary potential. How, for example, might a self-incompatible species fare as a small founder population or remnant population in a severe population bottleneck (see Concept 23.3), as compared with a self-fertile species?

10. **SCIENTIFIC INQUIRY** Critics of GM foods have argued that transgenes may disturb cellular functioning, causing unexpected and potentially harmful substances to appear inside cells. Toxic intermediary substances that normally occur in very small amounts may arise in larger amounts, or new substances may appear. The disruption may also lead to loss of substances that help maintain normal metabolism. If you were your nation's chief scientific advisor, how would you respond to these criticisms?

11. **SCIENCE, TECHNOLOGY, AND SOCIETY** Humans have engaged in genetic manipulation for millennia, producing plant and animal varieties through selective breeding and hybridization that significantly modify genomes of organisms. Why do you think modern genetic engineering, which often entails introducing or modifying only one or a few genes, has met with so much opposition? Should some forms of genetic engineering be of greater concern than others? Explain.

12. **WRITE ABOUT A THEME: ORGANIZATION** In a short essay (100–150 words), discuss how a flower's ability to reproduce with other flowers of the same species is an emergent property arising from floral parts and their organization.

13. **SYNTHESIZE YOUR KNOWLEDGE**

This colorized SEM shows pollen grains from six plant species. Explain how a pollen grain forms, how it functions, and how pollen grains contributed to the dominance of angiosperms and other seed plants.

For selected answers, see Appendix A.